Indian Statistical Institute, Kolkata

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IgAN: Overview and Pathologic Risk Stratification

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Contents

1 Introduction

1.1 Abstract

IgA Nephropathy (IgAN) is a disease where a protein from the immune system called IgA builds up in the kidney filters. Over time, this causes inflammation and makes the kidneys less able to filter waste from the blood properly. This can lead to blood in the urine, protein in the urine, high blood pressure, and gradual kidney damage. Some patients remain stable for a long time, while others slowly develop chronic kidney disease.

In European countries, IgAN has been studied extensively with considerably sized cohorts. In contrast, Indian cohorts are not extensively studied, so progression in this population is not well understood. Since the condition progresses unpredictably, it is difficult to predict progression risk easily. This paper aims to predict the risk factors for a cohort of Indian patients from a particular medical institution in Kolkata based on clinical and pathological variables.

1.2 Problem Statement

IgAN is a progressive kidney disease with highly variable clinical outcomes. While international risk-prediction tools exist, their applicability to Indian patient populations remains uncertain due to:

- Limited large-scale cohort studies in India.
- Differences in clinical presentation and progression patterns.
- Lack of external validation for existing prediction models in South Asian populations.

This analysis intends to:

1. Identify key clinical and histopathologic predictors of IgAN progression in an Indian cohort.
2. Apply a Cox proportional hazards modeling framework adapted from international studies.
3. Quantify the contribution of individual risk factors using feature attribution methods.

1.3 Data Overview

The dataset consists of patients diagnosed with primary IgAN from a medical institution in Kolkata.

Dataset characteristics:

- Total observations: 297 patients
- Final usable dataset after handling missing values: 71 patients
- Follow-up duration: Median 10 months (IQR 1.5–21; range 0–115 months)
- Progression events: 29 patients (40.85% of complete cases)

Variables include:

- **Clinical:** Age, Sex, Diabetes Mellitus, Hypertension, Dialysis status, Serum Creatinine, eGFR, Proteinuria, Hematuria
- **Pathology:** MEST-C scores (M, E, S, T, C)

1.4 Research Questions

- How well do clinical variables (eGFR, proteinuria, hypertension) predict IgAN progression in this Indian cohort?
- Does adding MEST-C pathology scores improve risk prediction beyond clinical variables alone?
- Which individual features (eGFR, hypertension, or proteinuria categories) contribute most to a patient's predicted progression risk?
- How do the findings from this Indian cohort compare to previously published international risk-prediction models?

1.5 Key Definitions

1.5.1 IgA Nephropathy (IgAN)

A disease where IgA immune complexes deposit in the glomerular mesangium, leading to inflammation, progressive kidney damage, and potential end-stage renal disease.

1.5.2 MEST-C Score (Oxford Classification)

A histopathologic scoring system used in IgAN to summarize biopsy features associated with renal outcomes:

- **M - Mesangial hypercellularity:** Too many cells in the kidney filter support area.
- **E - Endocapillary hypercellularity:** The tiny filter channels appear crowded or inflamed.

- **S - Segmental glomerulosclerosis:** Part of a kidney filter has developed scarring.
- **T - Tubular atrophy / interstitial fibrosis:** Kidney tubes and nearby tissue show long-term damage or scarring.
- **C - Cellular or fibrocellular crescents:** Crescent-shaped injury suggesting active or recent severe inflammation.

1.5.3 eGFR (Estimated Glomerular Filtration Rate)

Calculated using the CKD-EPI 2021 equation without the race factor:

$$\mathrm{eGFR} = 142 \times \min\left(\frac{\mathrm{Scr}}{\kappa}, 1\right)^{\alpha} \times \max\left(\frac{\mathrm{Scr}}{\kappa}, 1\right)^{-1.200} \times 0.9938^{\mathrm{Age}} \times 1.012^{\mathrm{female}}$$

Where:

- Scr: serum creatinine
- $\kappa = 0.7$ for females, 0.9 for males
- $\alpha = -0.241$ for females, -0.302 for males

1.5.4 Progression (Outcome Variable)

Defined as:

$$\text{Progression} = \begin{cases} 1, & \text{if eGFR decrease was } > 50\% \text{ or LRRAR/dialysis was mentioned} \\ 0, & \text{otherwise} \end{cases}$$

1.5.5 Hazard Ratio (HR)

Compares the instantaneous risk of progression between two groups or between different values of a predictor. HR = 1 indicates no difference; HR $>$ 1 indicates higher hazard; HR $<$ 1 indicates lower hazard.

2 Data Description and Cleaning

2.1 The Dataset

The initial dataset is obtained from a renal care institute in Kolkata. It consists of 290 patients diagnosed with IgaN. However, there were many cases where an important set of variables are missing(and in some cases the entire set of variables) or the patients did not follow up *ie* only 1 visit.

2.2 Data Cleaning

This led to a tedious data cleaning procedure, reducing the data size to 89 observations. 1 visit observations were only included if the patient had to undergo dialysis or LRRAR at that time. The variables included where Age, Sex, Initial Date, Final Date(of visit or event explained later), duration, hematuria, proteinuria level, creatinine level(initial and final), egfr(initial and final), dm(diabetes mellitus), htn(hypertension), mestc score and event.

Detailed description of variables

1. Age: Age of the individual in years at the time of first visit
2. Sex: Male(0) or Female(1)
3. DM: Yes(1) or No(0)
4. HTN: Yes(1) or No(0)
5. Hematuria: Yes(1) or No(0). 1 is assigned when more than 2 RBCs/hpf was reported. In case of ranges, if the median lied above 2, 1 was assigned.
6. Proteinuria: 5 levels were assigned and 4 columns(1,2,3,4) were made to note. The initial data had different notations like "protein++" or "x mg/gm". The following describes the assignment rules:
   - 0 if "negative" or "trace"
   - 1 if "present(+)" or "¡30mg/gm"
   - 2 if "present(++)" or "100-299mg/dL"
   - 3 if "present(+++)" or "300-999mg/dL"
   - 4 if "present(++++)" or ">999mg/dL"
7. Efgr: Egfr was calculated using the creatinine levels given at different times. The CKD-EPI 2021 equation was used for that purpose which is as follows:

   $$\text{eGFR} = 142 \times \min\left(\frac{\text{Scr}}{\kappa}, 1\right)^{\alpha} \times \max\left(\frac{\text{Scr}}{\kappa}, 1\right)^{-1.200} \times 0.9938^{\text{Age}} \times 1.012^{\text{female}}$$

   Where:

   - Scr: serum creatinine
   - $\kappa = 0.7$ for females, 0.9 for males
   - $\alpha = -0.241$ for females, -0.302 for males

8. Event: Event(Progression) was assigned 1 or 0 using by observing the egfr values or any mention of dialysis or LRRAR.

$$\text{Progression} = \begin{cases} 1, & \text{if eGFR decrease was } > 50\% \text{ or LRRAR/dialysis was mentioned} \\ 0, & \text{otherwise} \end{cases}$$

9. Date: The initial date was always noted as the first date where the patient reports all the other variables(creatinine, hematuria, proteinuria). The final date was noted as the earlier of the reporting date where dialysis/LRRAR is mentioned or the first reporting date where egfr declined from initial value by $> 50\%$ or the final reporting date if no event occured.

10. Duration: It was noted as the number of months between Initial and Final dates. Lower ceiling was used.

11. MESTC score: Corresponding scores as per Oxford Classifications. These were not reported seperately consistently for all 89 patients. In some cases it had to be obtained from the kidney biopsy report. However in several instances, as the total number of sampled glomeruli was less than 7, MESTC score was not even applicable as per the Oxford guidelines which further reduced datapoints with this variable.

3 Exploratory Data Analysis

3.1 Gender Distribution

Figure 1 shows the gender distribution of the 89 patients. About two-thirds of the patients were male.

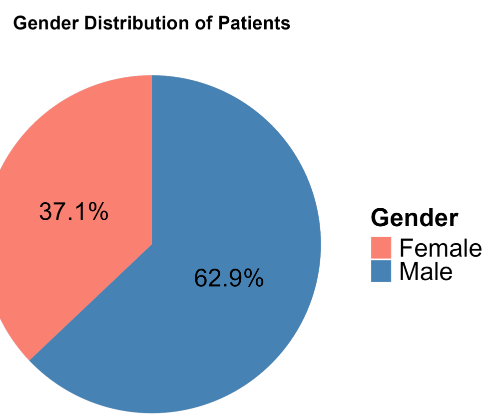


Figure 1: Gender distribution of patients.

3.2 Age Group Distribution

Figure 2 presents the bar chart. No patients were found below the age of 12 or above the age of 72. The mean age of first time visit is 38.68 and we see almost half of the data is concentrated in 26-45 age group.

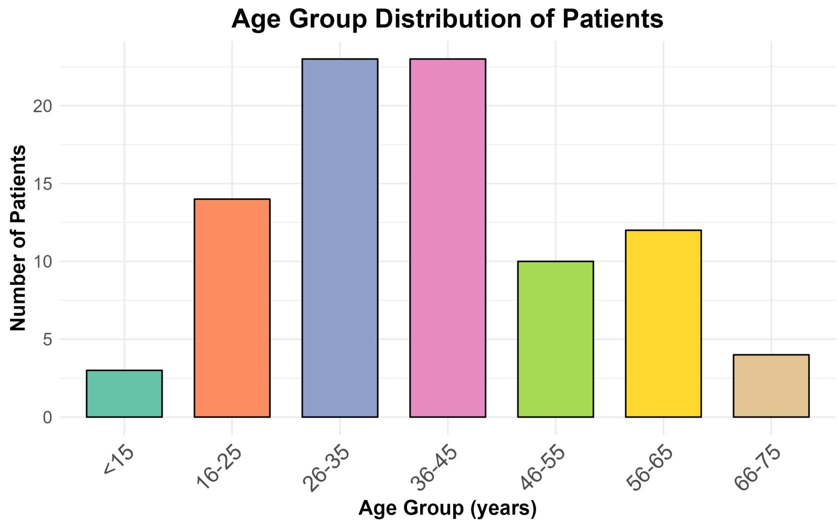
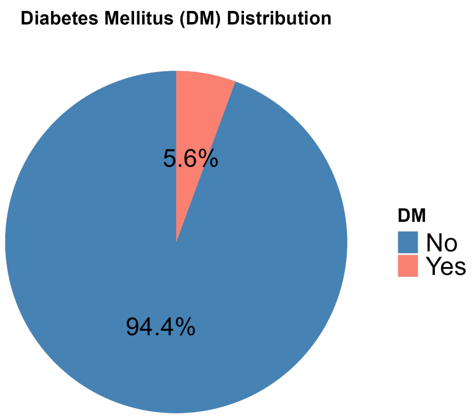


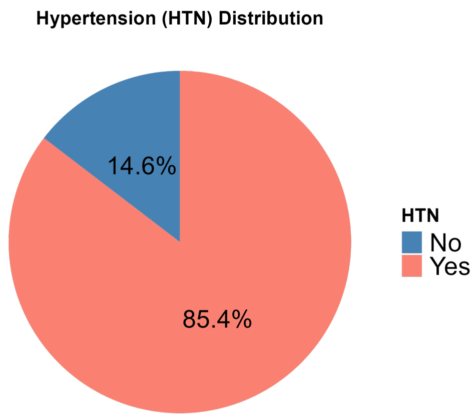
Figure 2: Age group distribution (years).

3.3 Comorbidities: Diabetes Mellitus and Hypertension

Figures 3a and 3b display the prevalence of DM and HTN. DM was present in only 5 out of 89, whereas we see a stark contrast in the prevalence if high blood presssure in the population with only 12 out of 89 reporting negative.



(a) Diabetes mellitus.



(b) Hypertension.

Figure 3: Prevalence of comorbidities.

3.4 Hematuria

Hematuria (presence of blood in urine) was recorded as a binary variable. Figure 4 shows that hematuria was present in 62.9% of the cohort's first report.

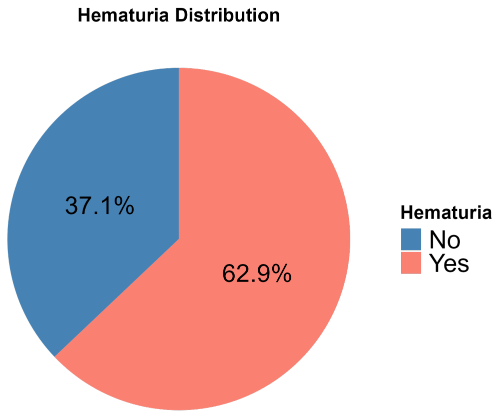


Figure 4: Hematuria distribution.

3.5 Proteinuria Levels

Proteinuria was graded on a scale from 0 (none) to 4 (most severe). Figure 5 illustrates the distribution. Most patients had proteinuria level 2. We see an almost symmetric distribution with low frequency in both extremes.

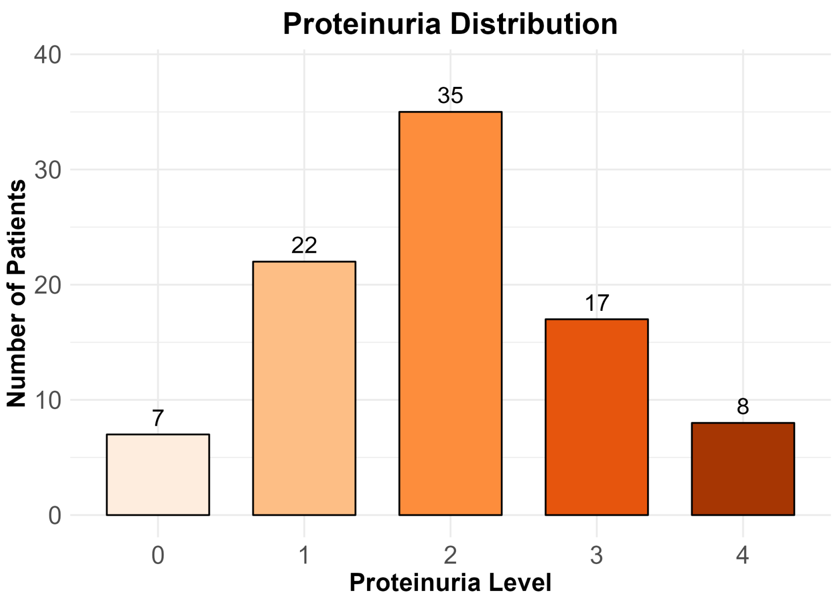
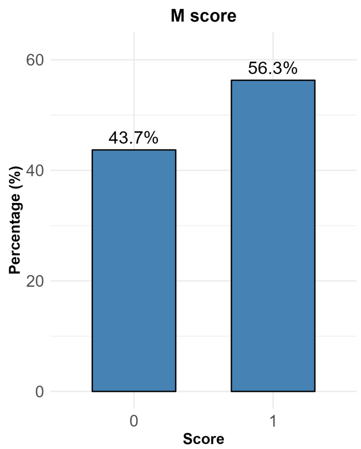


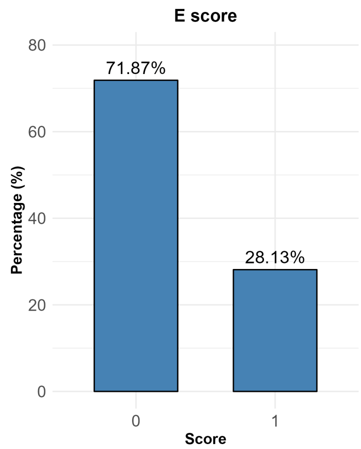
Figure 5: Proteinuria levels (0–4).

3.6 MEST-C Scores

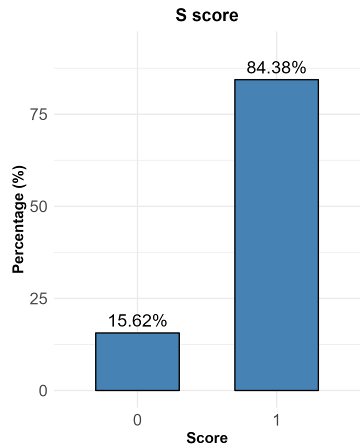
Figure 6 presents the frequency distribution of each MEST-C component. Slightly more than half of the patients (56.3%) had a positive M score, while nearly three-quarters (71.9%) were E0. The S score was positive (S1) in 84.4% of the cohort, making it the most common histopathological finding among these IgAN patients. For the T score, approximately one-tenth of patients (9.4%) had T2, while the remaining patients were almost evenly split between T0 (46.9%) and T1 (43.8%). Crescents (C score) were uncommon: only about one-quarter of patients (25.0%) had a positive C value, and among those, only a small fraction (9.4% of the total cohort) had C2.



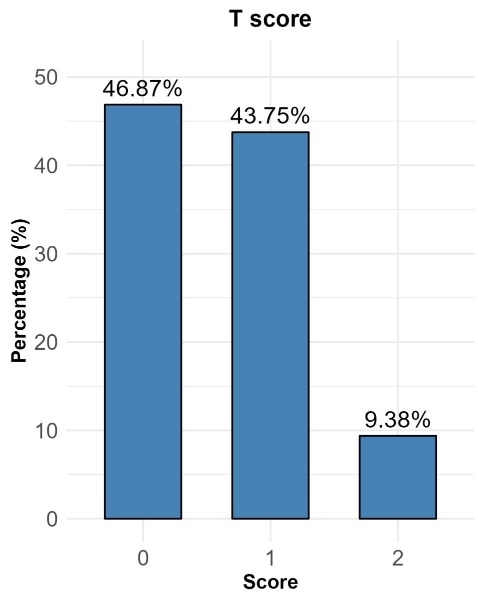
(a) M (mesangial hypercellularity).



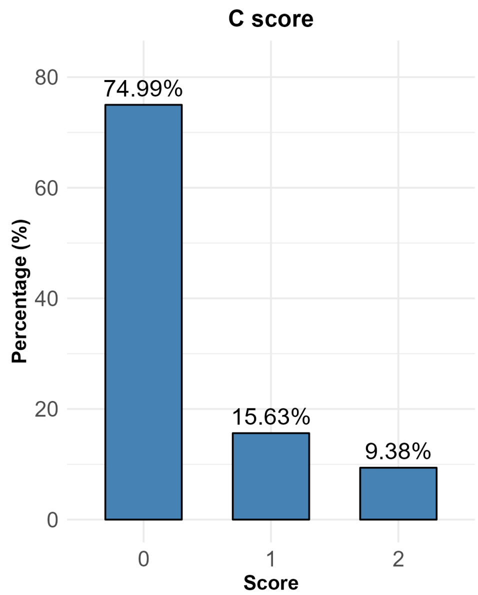
(b) E (endocapillary proliferation).



(c) S (segmental sclerosis).



(d) T (interstitial fibrosis/tubular atrophy).



(e) C (crescents).

Figure 6: Distribution of MEST-C scores.

3.7 Outcome Event

The primary endpoint (event) occurred in 37 patients (41.6%). Figure 7 shows the event distribution..

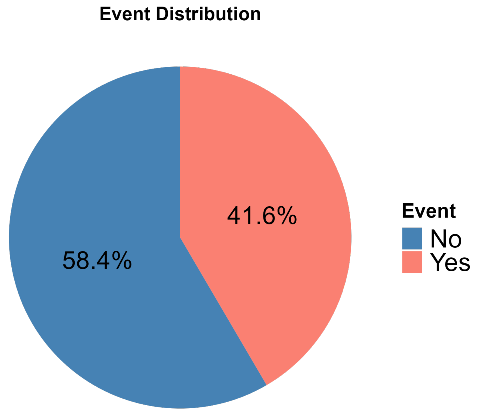


Figure 7: Occurrence of the composite event.

4 Cox Proportional Hazards Model

4.1 Introduction to Survival Analysis

Since IgA nephropathy progression occurs over varying follow-up durations and some patients may not experience progression during the observation period, survival analysis techniques were used. Survival analysis allows modeling of time-to-event outcomes while accounting for censored observations.

In this study, the event of interest was renal disease progression, defined as either:

- more than 50% decline in eGFR,
- dialysis initiation, or
- LRRAR mention in clinical records.

Patients without progression during follow-up were treated as censored observations.

4.2 Why Cox Proportional Hazards Model Was Used

The Cox proportional hazards model was selected because:

- it accommodates censored survival data,
- it does not require specification of the baseline hazard function,

- it allows estimation of the effect of multiple clinical and pathological predictors on progression risk simultaneously,
- and it is widely used in nephrology and medical survival-analysis research.

The modeling framework was adapted from the international IgA nephropathy risk-prediction study by Barbour et al. (2019).

4.3 Mathematical Formulation

The Cox proportional hazards model estimates the hazard of progression at time t for a patient with predictor vector X.

$$h(t|X) = h_0(t)e^{\beta X}$$

where:

- $h(t|X)$ denotes the hazard function conditional on predictors,
- $h_0(t)$ represents the baseline hazard function,
- β denotes the regression coefficients,
- X represents the predictor variables included in the model.

The Cox proportional hazards model assumes that hazard ratios remain constant over time. This assumption is known as the proportional hazards assumption and implies that the effect of each predictor on the hazard remains multiplicative and time-invariant throughout the follow-up period.

4.4 Hazard Ratio Interpretation

The regression coefficients obtained from the Cox proportional hazards model were interpreted using hazard ratios (HR), calculated as:

$$HR = e^{\beta}$$

The hazard ratio measures the relative instantaneous risk of progression associated with a predictor variable while holding other variables constant.

An HR greater than 1 indicates increased progression risk, whereas an HR less than 1 indicates reduced progression risk. An HR equal to 1 indicates no difference in hazard between groups.

4.5 Clinical Cox Model

The first survival model included only clinical variables available in the dataset. The model specification was:

$$Surv(time, event) = eGFR + Proteinuria + HTN$$

where:

- eGFR represents baseline estimated glomerular filtration rate,
- Proteinuria represents categorized urine protein levels,
- and HTN denotes hypertension status.

Proteinuria information was available as semi-quantitative dipstick categories recorded using "+" grading. Separate binary variables (Prot1–Prot4) were created to represent increasing levels of proteinuria severity corresponding to 1+, 2+, 3+, and 4+ proteinuria, respectively. Entries coded as 0 represented unavailable or missing proteinuria information rather than absence of proteinuria. So, our revised model specification was :

$$Surv(time, event) = eGFR + Proteinuria1 + Proteinuria2 + Proteinuria3 + Proteinuria4 + HTN$$

The clinical model was fitted using 88 complete observations.

4.6 Result of Clinical Model:

A Cox proportional hazards regression model was fitted using time-to-event data with the following predictors: initial eGFR, hypertension (HTN), and four proteinuria indicators.

The fitted model was

$$h(t|X) = h_0(t) \exp\left(\beta_1 \text{eGFR} + \beta_2 \text{HTN} + \beta_3 \text{prot1} + \beta_4 \text{prot2} + \beta_5 \text{prot3} + \beta_6 \text{prot4}\right)$$

The sample consisted of $n = 88$ patients with 36 observed events.

4.6.1 Model Summary

4.6.2 Interpretation

- eGFR had a significant protective effect on survival. For every one-unit increase in eGFR, the hazard decreased by approximately

$$(1 - 0.927) \times 100 \approx 7.3\%$$

Variable	Coef	HR	SE	z	p-value	95% CI for HR
eGFR	-0.0755	0.927	0.0151	-5.016	5.29×10^{-7}	(0.900, 0.955)
HTN=1	-2.2111	0.110	0.7992	-2.767	0.0057	(0.023, 0.525)
prot1=1	3.2327	25.348	1.2094	2.673	0.0075	(2.369, 271.265)
prot2=1	2.3799	10.804	1.1220	2.121	0.0339	(1.198, 97.420)
prot3=1	2.9101	18.358	1.1704	2.486	0.0129	(1.852, 182.015)
prot4=1	1.6492	5.203	1.0797	1.527	0.1267	(0.627, 43.181)

Table 1: Cox proportional hazards regression results

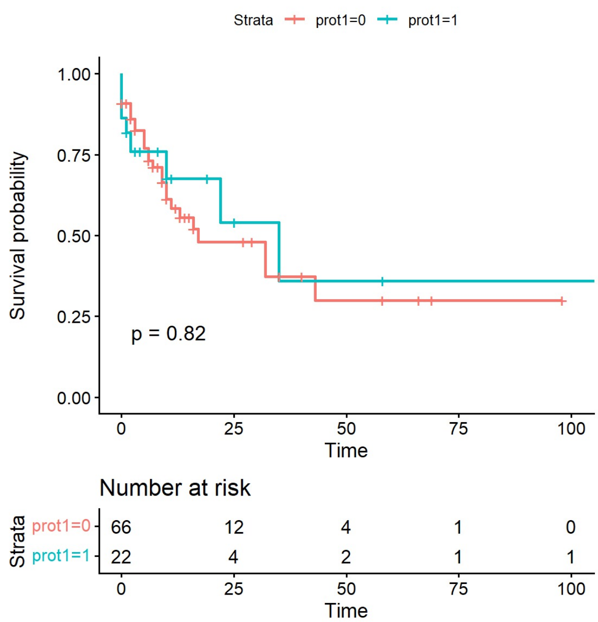
holding other variables constant.

- Hypertension (HTN=1) was associated with a significantly lower hazard compared to HTN=0:

$$HR = 0.11, \quad p = 0.0057$$

- Proteinuria variables prot1, prot2, and prot3 were significantly associated with increased hazard.
- prot4 was not statistically significant ($p = 0.127$).

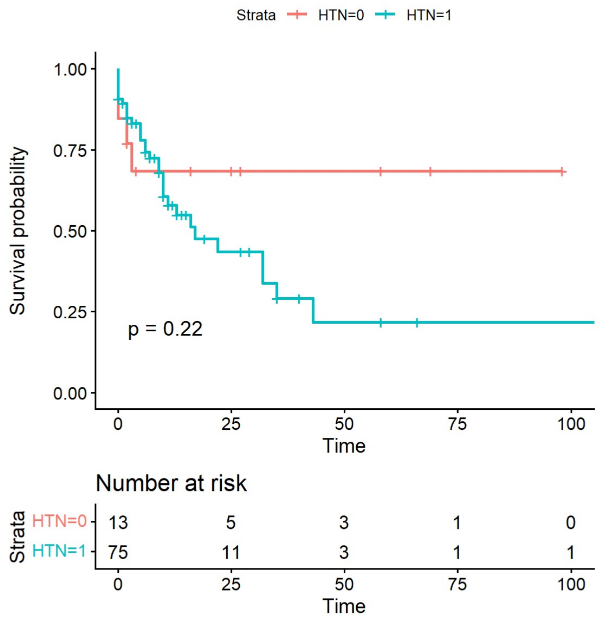
Figure 8: Kaplan–Meier Survival Analysis for prot1



Kaplan–Meier Survival Analysis for prot1

Kaplan–Meier survival analysis showed no statistically significant difference in survival between the two `prot1` groups (log-rank test: $p = 0.82$). The survival curves largely overlapped throughout the follow-up period, indicating similar survival experiences between the groups.

Figure 9: Kaplan–Meier Survival Analysis for HTN



Kaplan–Meier Survival Analysis for HTN

Kaplan–Meier analysis demonstrated comparatively lower survival probabilities among hypertensive patients (`HTN=1`) than non-hypertensive patients (`HTN=0`); however, the difference was not statistically significant (log-rank test: $p = 0.22$).

Overall Interpretation

Neither `prot1` nor `HTN` showed a statistically significant association with survival in the univariate Kaplan–Meier analysis.

Overall Model Performance

- Concordance index:

$$C = 0.841$$

with standard error

$$SE(C) = 0.0339$$

indicating good predictive discrimination.

- Likelihood ratio test:

$$\chi^2 = 50.55, \quad df = 6, \quad p < 0.001$$

- Wald test:

$$\chi^2 = 26.53, \quad df = 6, \quad p < 0.001$$

- Score (log-rank) test:

$$\chi^2 = 27.00, \quad df = 6, \quad p < 0.001$$

4.6.3 Explained Randomness

Using the concordance-based explained randomness measure,

$$\kappa = \sqrt{\frac{8}{\pi}}$$

$$D = \Phi^{-1}(C)\kappa$$

$$R_D^2 = \frac{D^2/\kappa^2}{\sigma^2 + D^2/\kappa^2}$$

where

$$\sigma^2 = \frac{\pi^2}{6}$$

the estimated explained randomness was

$$R_D^2 = 0.377$$

indicating that approximately 37.7% of variation in survival outcome was explained by the model.

Model performance was additionally evaluated using the Akaike Information Criterion (AIC). Also, the AIC for the clinical model is

$$AIC = 232.333$$

Proportional Hazards Assumption

The proportional hazards assumption was assessed using Schoenfeld residuals.

4.6.4 First PH Test

Variable	Chi-square	df	p-value
eGFR	0.513	1	0.4739
HTN	7.450	1	0.0063
prot1	0.00006	1	0.9938
prot2	0.004	1	0.9468
prot3	2.710	1	0.0994
prot4	0.399	1	0.5279
GLOBAL	14.80	6	0.0218

Table 2: Schoenfeld residual test for proportional hazards

4.6.5 Extended PH Test

Variable	Chi-square	df	p-value
age	0.209	1	0.6476
sex	0.358	1	0.5498
hematuria	0.003	1	0.9586
eGFR	0.503	1	0.4782
HTN	7.231	1	0.0072
prot1	0.192	1	0.6617
prot2	0.022	1	0.8823
prot3	4.504	1	0.0338
prot4	0.421	1	0.5166
GLOBAL	17.950	9	0.0358

Table 3: Extended proportional hazards test

4.6.6 Conclusion

The Cox regression model demonstrated strong predictive performance with a concordance index of 0.841. Lower eGFR and positive proteinuria indicators were associated with increased hazard. However, the proportional hazards assumption may be violated for HTN and prot3, as indicated by significant Schoenfeld residual tests.

4.7 Limited Cox Model Including MEST Variables

To evaluate whether biopsy-based pathological features improve progression-risk prediction, a second Cox model was fitted by incorporating Oxford MEST variables along with clinical predictors.

The model specification was:

$$Surv(time, event) = eGFR + Proteinuria + HTN + M + E + S + T$$

where:

- M represents mesangial hypercellularity,
- E represents endocapillary hypercellularity,
- S represents segmental glomerulosclerosis,
- and T represents tubular atrophy/interstitial fibrosis.

Proteinuria information was available as semi-quantitative dipstick categories recorded using "+" grading. Separate binary variables (Prot1–Prot4) were created to represent increasing levels of proteinuria severity corresponding to 1+, 2+, 3+, and 4+ proteinuria, respectively. Entries coded as 0 represented unavailable or missing proteinuria information rather than absence of proteinuria. The tubular atrophy/interstitial fibrosis component of the Oxford MEST-C classification was represented using separate binary indicator variables corresponding to T1 and T2 lesion categories. T1 represented moderate tubular atrophy/interstitial fibrosis (26–50%), while T2 represented severe fibrosis (¿50%). So, our revised model specification was :

$$Surv(time, event) = eGFR + Proteinuria1 + Proteinuria2+$$
$$Proteinuria3 + Proteinuria4 + HTN + M + E + S + T1 + T2$$

The Limited model was fitted using 32 complete observations.

4.8 Result of Limited Model

A limited multivariable Cox proportional hazards regression model was fitted using baseline clinical variables including eGFR, hypertension status, and proteinuria indicators.

The analysis included $n = 88$ patients with 36 observed events.

4.9 Model Performance

The overall Cox regression model was statistically significant:

$$\chi^2_{LR} = 50.55, \quad df = 6, \quad p < 0.001$$

The concordance index of the model was

$$C = 0.841 \pm 0.034$$

indicating good discriminative ability.

The explained randomness measure based on the concordance index was

$$R_D^2 = 0.377$$

Model performance was additionally evaluated using the Akaike Information Criterion (AIC). Also, the AIC for the clinical model is

$$AIC = 50.11864$$

suggesting that approximately 37.7% of the variability in survival outcome was explained by the model.

4.10 Hazard Ratios

Variable	Hazard Ratio	95% CI	p-value
eGFR	0.927	(0.900, 0.955)	< 0.001
HTN=1	0.110	(0.023, 0.525)	0.0057
prot1=1	25.35	(2.37, 271.27)	0.0075
prot2=1	10.80	(1.20, 97.42)	0.0339
prot3=1	18.36	(1.85, 182.01)	0.0129
prot4=1	5.20	(0.63, 43.18)	0.1267

Table 4: Limited Cox proportional hazards regression model

4.11 Interpretation

Higher baseline eGFR was associated with a significantly lower hazard of progression:

$$HR = 0.927, \quad p < 0.001$$

indicating an approximately 7.3% reduction in hazard for every one-unit increase in eGFR.

Hypertension was significantly associated with lower hazard:

$$HR = 0.11, \quad p = 0.0057$$

Proteinuria categories prot1, prot2, and prot3 were significantly associated with increased hazard, while prot4 did not reach statistical significance.

4.12 Assessment of Proportional Hazards Assumption

The proportional hazards assumption was evaluated using Schoenfeld residuals.

Variable	Chi-square	df	p-value
eGFR	0.513	1	0.474
HTN	7.450	1	0.006
prot1	0.00006	1	0.994
prot2	0.004	1	0.947
prot3	2.710	1	0.099
prot4	0.399	1	0.528
GLOBAL	14.80	6	0.022

Table 5: Schoenfeld residual test for proportional hazards

The proportional hazards assumption appeared reasonable for most covariates. However, HTN showed evidence of violation of the proportional hazards assumption ($p = 0.006$). The global test was also statistically significant ($p = 0.022$), suggesting possible non-proportionality in the model.

5 Integrated Gradients: A Feature Attribution Method

Intuition

Imagine a model that predicts a person's credit score. For one applicant, the prediction is much higher than average.

The model took several inputs: income, loan history, debt ratio, employment status.

Attribution decomposes the final prediction into:

$$\text{Prediction} = \text{Baseline} + \sum_i \text{Contribution}_i$$

This tells us why this specific applicant received that score, not just a global trend. Also the AIC for the limited model is 50.11864.

Definition

Integrated Gradients (IG) is an attribution method that explains a model's prediction for a given input by accumulating the gradients of the model's output along a straight-line path from a baseline input to the actual input.

5.1 Attribution Formula

For a model

$$F : R^n \to R,$$

an input

$$x = (x_1, \dots, x_n),$$

and a baseline

$$x' = (x'_1, \dots, x'_n),$$

the IG attribution for feature i is:

$$IG_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F\big(x' + \alpha(x - x')\big)}{\partial x_i}\, d\alpha$$

5.2 Important Property

Integrated Gradients satisfies completeness (summation to delta):

$$\sum_{i=1}^{n} IG_i(x) = F(x) - F(x')$$

5.3 Integrated Gradients for the Clinical Model

The clinical Cox model uses the linear predictor

$$\eta(x) = \beta_{\mathrm{eGFR}} x_{\mathrm{eGFR}} + \beta_{\mathrm{Prot1}} x_{\mathrm{Prot1}} + \beta_{\mathrm{Prot2}} x_{\mathrm{Prot2}} + \beta_{\mathrm{Prot3}} x_{\mathrm{Prot3}} + \beta_{\mathrm{HTN}} x_{\mathrm{HTN}}$$

All partial derivatives are constant, so the IG integral simplifies to:

$$IG_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial \eta}{\partial x_i}\, d\alpha = \beta_i (x_i - x'_i)$$

5.4 Feature Attributions

$$IG_{\mathrm{eGFR}} = \beta_{\mathrm{eGFR}} (x_{\mathrm{eGFR}} - x'_{\mathrm{eGFR}})$$

$$IG_{\mathrm{HTN}} = \beta_{\mathrm{HTN}} (x_{\mathrm{HTN}} - x'_{\mathrm{HTN}})$$

$$IG_{\mathrm{Prot1}} = \beta_{\mathrm{Prot1}} (x_{\mathrm{Prot1}} - x'_{\mathrm{Prot1}})$$

$$IG_{\mathrm{Prot2}} = \beta_{\mathrm{Prot2}} (x_{\mathrm{Prot2}} - x'_{\mathrm{Prot2}})$$

$$IG_{\mathrm{Prot3}} = \beta_{\mathrm{Prot3}} (x_{\mathrm{Prot3}} - x'_{\mathrm{Prot3}})$$

5.5 Baseline and Coefficients

Baseline values:

$$x'_{\text{eGFR}} = 90, \qquad x'_{\text{HTN}} = 0, \qquad x'_{\text{Prot1}} = x'_{\text{Prot2}} = x'_{\text{Prot3}} = x'_{\text{Prot4}} = 0$$

Coefficients (log hazard ratios):

$$\beta_{\text{eGFR}} = \ln(0.92729) \approx -0.07549$$

$$\beta_{\text{HTN}} = \ln(0.10958) \approx -2.211$$

$$\beta_{\text{Prot1}} = \ln(25.34839) \approx 3.2327$$

$$\beta_{\text{Prot2}} = \ln(10.80353) \approx 2.3799$$

$$\beta_{\text{Prot3}} = \ln(18.35822) \approx 2.91$$

$$\beta_{\text{Prot4}} = \ln(5.20258) \approx 1.65$$

IG attributions (general form):

$$IG_{\text{eGFR}} = -0.07549 \times (x_{\text{eGFR}} - 90)$$

$$IG_{\text{HTN}} = -2.211 \times (x_{\text{HTN}} - 0)$$

$$IG_{\text{Prot1}} = 3.2327 \times (x_{\text{Prot1}} - 0)$$

$$IG_{\text{Prot2}} = 2.3799 \times (x_{\text{Prot2}} - 0)$$

$$IG_{\text{Prot3}} = 2.91 \times (x_{\text{Prot3}} - 0)$$

$$IG_{\text{Prot4}} = 1.65 \times (x_{\text{Prot4}} - 0)$$

These formulas decompose any patient's risk shift relative to a "healthy" baseline.

Now we calculate IG for 5 random patients from the dataset

Interpretation of Integrated Gradient Attributions

The Integrated Gradient (IG) attributions quantify the contribution of each predictor variable to the model prediction for a patient.

- **eGFR** contribution is the largest in most of them perhaps because the eGFR levels deviate a lot from baseline(90) despite being weighted by its corresponding coefficient. This suggests eGFR strongly contributed towards the model prediction,

Table 6: Integrated Gradients attribution for five randomly selected patients. Baseline: eGFR = 90, HTN = 0, proteinuria = 0.

Patient	eGFR	HTN	Prot.	Event	IG (log-hazard)		
(ID)	(mL/min)	(Yes/No)	(grade 0-4)	(0/1)	eGFR	HTN	Prot.
pt196	51.5	No	1	1	2.91	0.00	3.23(prot 1)
pt155	9.81	Yes	2	1	6.05	-2.21	2.38(prot 2)
pt43	10.1	Yes	4	0	6.03	-2.21	1.65(prot 4)
pt9	86.1	No	2	0	0.30	0.00	2.38(prot 2)
pt208	134	Yes	0	0	-3.34	-2.21	0.00

- **HTN1** IG is 0 for patients who do not suffer from high blood pressure. For the ones with hypertension, we have IG as -2.21. This suggests HTN has an inverse effect(non occurence of event) im model prediction. .
- **prot1** (3.23), **prot2** (2.38), **prot3** (2.91), and **prot4** (1.65) show positive contribution, which is expected. However the magnitude is highest in 1 and lowest in 4. This may occur due to only 8 observations in prot4 which may undermine its effect.

6 Data Management Recommendation

Although the total number of observations was 292, the clinical model could only be fitted using 89 patients — less than one-third of the total. While this is partly explained by new patients having only one visit (insufficient data for progression analysis) or by patients with no follow-up visits, many of the remaining exclusions were avoidable. A larger dataset could have been retained. Furthermore, most variables had to be categorized because many observations recorded results in categories rather than as original continuous values or ranges

- Hematuria : Record as RBCs per HPF whenever possible, instead of "absent" or "present".
- Proteinuria : Multiple input formats were observed (e.g., "present(+)", "present(++)", or mg/g [ACR]). Reporting proteinuria as a quantity (rather than a category) would allow it to be treated as a continuous variable.
- MESTC score should be recorded whenever possible. There are instances where number of sampled glomeruli fails to meet oxford classification or the kidney biopsy report mentions inforamtion about certain parameters of MESTC but with not much data on the others. So MESTC score should definitely be extracted from biopsy reports whenever possible